

CHE 565 -- Homework 3

Soki Sem

April 18, 2026

Problem 1

Steepest Descent

Function :

$$f(x_1, x_2) = 0.5x_1^2 + 1.0x_1x_2 - x_1 + 1.0x_2^2 - x_2$$

At $x_0 = \text{Matrix}([1, 1])$, $f(x_0) = 0.5000000000000000$

Gradient of f:

$$\nabla f = \begin{bmatrix} 1.0x_1 + 1.0x_2 - 1.0 \\ 1.0x_1 + 2.0x_2 - 1 \end{bmatrix}$$

At $x_0 = \text{Matrix}([1, 1])$, $Df(x_0) = 1.00, 2.00$

$$\begin{bmatrix} g_1(\alpha) \\ g_2(\alpha) \end{bmatrix} = \begin{bmatrix} 1 - 1.0\alpha \\ 1 - 2.0\alpha \end{bmatrix}$$

Function ϕ_k :

$$\phi(\alpha) = 6.5\alpha^2 - 5.0\alpha + 0.5$$

Derivative of ϕ_k :

$$\begin{aligned} \frac{d}{d\alpha}\phi(\alpha) &= 13.0\alpha - 5.0 \\ 0 &= 13.0\alpha - 5.0 \end{aligned}$$

Optimal alpha: 0.38

Next iterate x_1 : 0.62, x_2 : 0.23

Function value at next iterate: -0.46

Problem 2

Conjugate Gradient Method

$$\begin{bmatrix} s_1(\alpha) \\ s_2(\alpha) \end{bmatrix} = \begin{bmatrix} 0.0946745562130178 \\ -0.195266272189349 \end{bmatrix}$$

Problem 3

Problem 3

Blending LP Formulation

Constituent	Max quantity (bbl/day)	Production Cost (\$/bbl)
1	3000	26.00
2	2000	30.60
3	4000	29.20
4	1000	29.80

Grade	Specifications	Selling Price (\$/bbl)
A	Not more than 15% of 1; Not less than 40% of 2	32.40
B	Not more than 50% of 3; Not more than 10% of 1	31.50
C	Not less than 10% of 2; Not more than 20% of 1	30.60

Decision variables :

InlineMathToken(latex ='
 $a_1,$
 $a_2,$
 $a_3,$
 a_4
 $right']'),$ *InlineMathToken(latex ='*
 $b_1,$
 $b_2,$
 $b_3,$
 b_4
 $right']'),$ *InlineMathToken(latex ='*
 $c_1,$
 $c_2,$
 $c_3,$
 c_4
 $right']')$

$$a_1 + b_1 + c_1 \leq 3000$$

$$a_2 + b_2 + c_2 \leq 2000$$

$$a_3 + b_3 + c_3 \leq 4000$$

$$a_4 + b_4 + c_4 \leq 1000$$

Total production:

$$A = a_1 + a_2 + a_3 + a_4$$

$$B = b_1 + b_2 + b_3 + b_4$$

$$C = c_1 + c_2 + c_3 + c_4$$

Objective (maximize profit):

$$P = 32.40A + 31.50B + 30.60C - [26(a_1 + b_1 + c_1) + 30.60(a_2 + b_2 + c_2) + 29.20(a_3 + b_3 + c_3) + 29.80(a_4 + b_4 + c_4)]$$

Subject to:

$$a_1 \leq 0.15A$$

$$a_2 \geq 0.40A$$

$$b_3 \leq 0.50B$$

$$b_1 \leq 0.10B$$

$$c_2 \geq 0.10C$$

$$c_1 \leq 0.20C$$

$$a_1 + b_1 + c_1 \leq 3000$$

$$a_2 + b_2 + c_2 \leq 2000$$

$$a_3 + b_3 + c_3 \leq 4000$$

$$a_4 + b_4 + c_4 \leq 1000$$

$$a_i, b_i, c_i \geq 0, \quad i = 1, \dots, 4$$